

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MTH 226 — Past Questions & Materials
by Renance

Course: MTH 226
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 2
Levels: 200

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Paper 1 of 2 — MTH 226 2022/2023

Level 200 • PostUTME • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCE:

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COLLEGE OF SCIENCE as

DEPARTMENT OF MATHEMATICS

SECOND SEMESTER EXAMINATION 2022/2023 SESSION

COURSE TITLE & CODE: ABSTRACT ALG

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Instruction: Attempt any four questions Time: 2 hours aE 229)

Question 1(17.5 marks)

- Define (i) Field (ii) Prime and Maximal Ideals (ii) Ring of a polynomial
- Prove that every field is an Integral domain .
- Prove that the characteristic of an integral domain is either zero or a prime number.
- Express $x^* + x^3 + 5x^? + x - 2as (x^? + 2x + 1)q(x) + r()$ in Zs where $q(x)$ is the polynomial of quotient and $r(x)$ the polynomial of remainder

Question 2 (17.5 marks)

- Define (i) Group (ii) Order of a group (iii) Subgroup
- Prove that the Identity and Inverse elements of a group are unique
- Prove directly from the definition of divisibility, that if $d|x$ and $d|y$, then $d|(x - y)$ and $d|(x + y)$.
- Show that the operation $*$ defined on the set of R of real numbers by $a * b = a + b + 2$ for all $a, b \in R$ is a group

Wend 3 (17.5 marks)

- Relatively prime Integers (ii) Common divisor and multiple (iii) Euclidean Algorithm
 - If $a|b$ and $b|c$, then $a|c$ (ii) If $d|x$ and $d|y$, then for any integers a, b we have $d|(ax + by)$.
- abelian group, where $G = \{(a,b): a, b \in R, a \neq 0\}$ and $*$ is defined on G by $(a,b) + (c,d) =$
- divisor of $(1236, 4323)$ and express it in the form $1236x + 4323y$

'osets (ii) Lagrange's theorem (iii) Monoid

of 2×2 over the complex numbers $Qg = \{+, \cdot, \cdot^4, \cdot^B, +C\}$ where $I = i a]. =$

$(5 \ 3]$ and $i = V - 1$. Show the distinct right cosets of the cyclic subgroup $H = (B)$.

$4Z$ in Z ?

$12) \in Sp$ (ii) $3 \in Zs$ (iil) $(ie \ s] \in a$

Integral Domain (iii) Ring

domain if and only if p is a prime number.

not Integral domains? $Z,, Zs, 2Z, Z + iZ, \{0\}$ and why? (ii) Draw the Cayley table

the elements of $Zz \times Z3$.

up (iii) Semigroup

(ii) that if H and K are subgroups of a group G . Then HK is also a

that $(xy)^n = y^n x^n$.

of the Integers $\mathbb{Z}$.

Paper 2 of 2 — MTH 226 PRATICE QUESTIONS

Level 200 • Second Semester • PastQuestion

MTH 226 PRATICE QUESTIONS

1. Define the following

- (a) Group
- (b) Order of a group
- (c) Subgroup
- (d) Cyclic group
- (e) Trivial subgroup
- (f) Semigroup
- (g) Zero divisors
- (h) Integral Domain
- (i) Ring
- (j) Left and Right Cosets
- (k) Lagrange's theorem
- (l) Relatively prime Integers
- (m) common divisor and multiple
- (n) Euclidean Algorithm
- (o) Monoid
- (p) Integral domain
- (q) Field
- (r) Prime and Maximal Ideals
- (s) Ring of a polynomial

2. Prove that the Identity and Inverse elements of a group are unique

3. Prove directly from the definition of divisibility, that if $\alpha|\beta$ and $\beta|\gamma$, then $\alpha|(\beta - \gamma)$ and $\alpha|(\beta + \gamma)$.

$\square\square$

4. Show that the operation $*$ defined on the set of $\mathbb{R}$ of real numbers by $\alpha * \beta = \alpha + \beta + 1$ for all $\alpha, \beta \in \mathbb{R}$ is a group

2

5. Prove (i) that every cyclic group is abelian (ii) that if H and K are subgroups of a group G . Then $H \cap K$ is also a subgroup of G .

6. Let x and y be elements of a group G . Prove that $(xy)^{-1} = y^{-1}x^{-1}$.

7. Show that the Even Integers $2\mathbb{Z}$ is a subgroup of the Integers $\mathbb{Z}$.

8. Prove the following (i) If $\alpha|\beta$ and $\beta|\gamma$, then $\alpha|\gamma$ (ii) If $\alpha|\beta$ and $\beta|\gamma$, then for any integers δ , $\delta\alpha$ we have $\delta\alpha|(\delta\beta + \delta\gamma)$.

9. Show that $(\mathbb{R}, *)$ is non-abelian group, where $\mathbb{R} = \{(\alpha, \beta): \alpha, \beta \in \mathbb{R}, \alpha \neq 0\}$ and $*$ is defined on G by $(\alpha, \beta) *$

$(\gamma, \delta) = (\alpha\gamma, \alpha\beta + \gamma\delta)$.

10. Find the greatest common divisor of (1236, 4323) and express it in the form $1236x + 4323y$

1 0

11. Given the following set of 2×2 over the complex numbers $\mathbb{C} = \{\pm 1, \pm i, \pm j, \pm k\}$ where $i = \sqrt{-1}$, $j = \sqrt{-1}$, $k = \sqrt{-1}$ and $i = \sqrt{-1}$. Show the distinct right cosets of the cyclic subgroup $H = \{1, i, -1, -i\}$.
12. What are the left cosets of $4\mathbb{Z}$ in $\mathbb{Z}$?
13. What are the orders of (i) $(1\ 2) \in S_3$ (ii) $3 \in \mathbb{Z}_4$ (iii) $[1] \in \mathbb{Z}_8$
14. Prove that $\mathbb{Z}_p$ is an Integral domain if and only if p is a prime number.
15. (i) Which of the following rings are not Integral domains? $\mathbb{Z}_4$, $\mathbb{Z}_5$, $2\mathbb{Z}$, $\mathbb{Z} + i\mathbb{Z}$, $\{0\}$ and why? (ii) Draw the Cayley table of addition and multiplication tables for the elements of $\mathbb{Z}_2 \times \mathbb{Z}_3$.
16. Prove that every field is an Integral domain.
17. Prove that the characteristic of an integral domain is either zero or a prime number.
18. Express $x^4 + x^3 + 5x^2 + x - 2$ as $(x^2 + 2x + 1)q(x) + r(x)$ in $\mathbb{Z}_3$ where $q(x)$ is the polynomial of quotient and $r(x)$ the polynomial of remainder